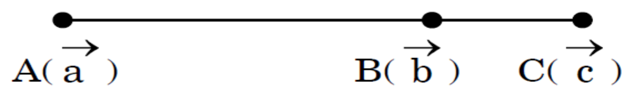
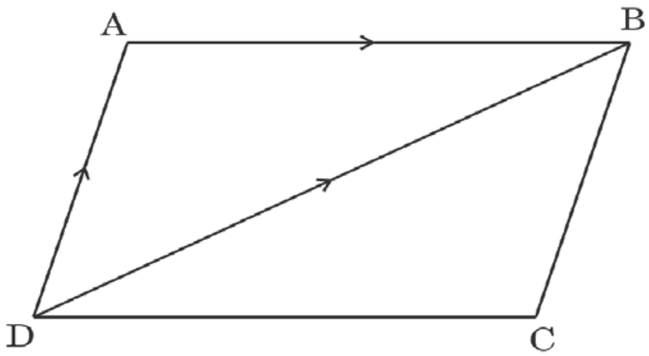
VECTORS

1.	The value of p for which the vectors $2\hat{i} + p\hat{j} + \hat{k}$ and $-4\hat{i} - 6\hat{j} + 26\hat{k}$ are perpendicular to each other, is : (a) 3 (b) -3 (c) $-\frac{17}{3}$ (d) $\frac{17}{3}$
2.	The value of $(\hat{i} \times \hat{j}) \cdot \hat{j} + (\hat{j} \times \hat{i}) \cdot \hat{k}$ is : (a) 2 (b) 0 (c) 1 (d) -1
3.	If $\vec{a} + \vec{b} = \hat{i}$ and $\vec{a} = 2\hat{i} - 2\hat{j} + 2\hat{k}$, then $\lvert \vec{b} \rvert$ equals : (a) $\sqrt{14}$ (b) 3 (c) $\sqrt{12}$ (d) $\sqrt{17}$
4.	Find all the vectors of magnitude $3\sqrt{3}$ which are collinear to vector $\hat{i} + \hat{j} + \hat{k}$.
5.	Position vectors of the points A, B and C as shown in the figure below are $\vec{a}$, $\vec{b}$ and $\vec{c}$ respectively.  If $\overrightarrow{AC} = \frac{5}{4}\overrightarrow{AB}$, express $\vec{c}$ in terms of $\vec{a}$ and $\vec{b}$.
6.	Unit vector along $\overrightarrow{PQ}$, where coordinates of P and Q respectively are (2, 1, –1) and (4, 4, –7), is (A) $2\hat{i} + 3\hat{j} - 6\hat{k}$ (B) $-2\hat{i} - 3\hat{j} + 6\hat{k}$ (C) $\frac{-2\hat{i}}{7} - \frac{3\hat{j}}{7} + \frac{6\hat{k}}{7}$ (D) $\frac{2\hat{i}}{7} + \frac{3\hat{j}}{7} - \frac{6\hat{k}}{7}$

7.	If in ΔABC, $\overrightarrow{BA} = 2\vec{a}$ and $\overrightarrow{BC} = 3\vec{b}$, then $\overrightarrow{AC}$ is (A) $2\vec{a} + 3\vec{b}$ (B) $2\vec{a} - 3\vec{b}$ (C) $3\vec{b} - 2\vec{a}$ (D) $-2\vec{a} - 3\vec{b}$
8.	If $\lvert \vec{a} \times \vec{b} \rvert = \sqrt{3}$ and $\vec{a} \cdot \vec{b} = -3$, then angle between $\vec{a}$ and $\vec{b}$ is (A) $\frac{2\pi}{3}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{5\pi}{6}$
9.	If $\vec{a}, \vec{b}, \vec{c}$ are three non-zero unequal vectors such that $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}$, then find the angle between $\vec{a}$ and $\vec{b} - \vec{c}$.
10.	If the angle between the vectors $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{4}$ and $\lvert \vec{a} \times \vec{b} \rvert = 1$, then $\vec{a} \cdot \vec{b}$ is equal to (a) -1 (b) 1 (c) $\frac{1}{\sqrt{2}}$ (d) $\sqrt{2}$
11.	$\vec{a}$ and $\vec{b}$ are two non-zero vectors such that the projection of $\vec{a}$ on $\vec{b}$ is 0. The angle between $\vec{a}$ and $\vec{b}$ is : (a) $\frac{\pi}{2}$ (b) π (c) $\frac{\pi}{4}$ (d) 0
12.	In Δ ABC, $\overrightarrow{AB} = \hat{i} + \hat{j} + 2\hat{k}$ and $\overrightarrow{AC} = 3\hat{i} - \hat{j} + 4\hat{k}$. If D is mid-point of BC, then vector $\overrightarrow{AD}$ is equal to : (a) $4\hat{i} + 6\hat{k}$ (b) $2\hat{i} - 2\hat{j} + 2\hat{k}$ (c) $\hat{i} - \hat{j} + \hat{k}$ (d) $2\hat{i} + 3\hat{k}$
13.	If $\vec{r} = 3\hat{i} - 2\hat{j} + 6\hat{k}$, find the value of $(\vec{r} \times \hat{j}) \cdot (\vec{r} \times \hat{k}) - 12$.

14.	A unit vector along the vector $4\hat{i} - 3\hat{k}$ is : (a) $\frac{1}{7}(4\hat{i} - 3\hat{k})$ (b) $\frac{1}{5}(4\hat{i} - 3\hat{k})$ (c) $\frac{1}{\sqrt{7}}(4\hat{i} - 3\hat{k})$ (d) $\frac{1}{\sqrt{5}}(4\hat{i} - 3\hat{k})$
15.	If θ is the angle between two vectors $\vec{a}$ and $\vec{b}$, then $\vec{a} \cdot \vec{b} \geq 0$ only when : (a) $0 < \theta < \frac{\pi}{2}$ (b) $0 \leq \theta \leq \frac{\pi}{2}$ (c) $0 < \theta < \pi$ (d) $0 \leq \theta \leq \pi$
16.	If the projection of the vector $\hat{i} + \hat{j} + \hat{k}$ on the vector $p\hat{i} + \hat{j} - 2\hat{k}$ is $\frac{1}{3}$, then find the value(s) of p.
17.	Two vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ are collinear if (a) $a_1b_1 + a_2b_2 + a_3b_3 = 0$ (b) $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$ (c) $a_1 = b_1, a_2 = b_2, a_3 = b_3$ (d) $a_1 + a_2 + a_3 = b_1 + b_2 + b_3$
18.	A unit vector $\hat{a}$ makes equal but acute angles on the co-ordinate axes. The projection of the vector $\hat{a}$ on the vector $\vec{b} = 5\hat{i} + 7\hat{j} - \hat{k}$ is (a) $\frac{11}{15}$ (b) $\frac{11}{5\sqrt{3}}$ (c) $\frac{4}{5}$ (d) $\frac{3}{5\sqrt{3}}$
19.	If the vectors $\vec{a}$ and $\vec{b}$ are such that $\|\vec{a}\| = 3$, $\|\vec{b}\| = \frac{2}{3}$ and $\vec{a} \times \vec{b}$ is a unit vector, then find the angle between $\vec{a}$ and $\vec{b}$.
20.	Find the area of a parallelogram whose adjacent sides are determined by the vectors $\vec{a} = \hat{i} - \hat{j} + 3\hat{k}$ and $\vec{b} = 2\hat{i} - 7\hat{j} + \hat{k}$.

21.	Three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ satisfy the condition $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. Evaluate the quantity $\mu = \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$, if $\lvert \vec{a} \rvert = 3$, $\lvert \vec{b} \rvert = 4$ and $\lvert \vec{c} \rvert = 2$.
22.	What is the value of $\dfrac{\text{projection of } \vec{a} \text{ on } \vec{b}}{\text{projection of } \vec{b} \text{ on } \vec{a}}$ for vectors $\vec{a} = 2\hat{i} - 3\hat{j} - 6\hat{k}$ and $\vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}$? (A) $\frac{3}{7}$ (B) $\frac{7}{3}$ (C) $\frac{4}{3}$ (D) $\frac{4}{7}$
23.	If $\vec{a}$ and $\vec{b}$ are two vectors such that $\vec{a} . \vec{b} > 0$ and $\lvert \vec{a} . \vec{b} \rvert = \lvert \vec{a} \times \vec{b} \rvert$, then the angle between $\vec{a}$ and $\vec{b}$ is : (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{2\pi}{3}$ (D) $\frac{3\pi}{4}$
24.	If the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ represent the three sides of a triangle, then show that $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$.
25.	If $\vec{a}$, $\vec{b}$ and $(\vec{a} + \vec{b})$ are all unit vectors and θ is the angle between $\vec{a}$ and $\vec{b}$, then the value of θ is : (a) $\frac{2\pi}{3}$ (b) $\frac{5\pi}{6}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$
26.	The projection of vector $\hat{i}$ on the vector $\hat{i} + \hat{j} + 2\hat{k}$ is : (a) $\frac{1}{\sqrt{6}}$ (b) $\sqrt{6}$ (c) $\frac{2}{\sqrt{6}}$ (d) $\frac{3}{\sqrt{6}}$
27.	If ABCD is a parallelogram and AC and BD are its diagonals, then $\overrightarrow{AC} + \overrightarrow{BD}$ is : (a) $2\,\overrightarrow{DA}$ (b) $2\,\overrightarrow{AB}$ (c) $2\,\overrightarrow{BC}$ (d) $2\,\overrightarrow{BD}$
28.	If three non-zero vectors are $\vec{a}$, $\vec{b}$ and $\vec{c}$ such that $\vec{a} . \vec{b} = \vec{a} . \vec{c}$ and $\vec{a} \times \vec{b} = \vec{a} \times \vec{c}$, then show that $\vec{b} = \vec{c}$.

29.	If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three vectors such that $\lvert\vec{a}\rvert = 7$, $\lvert\vec{b}\rvert = 24$, $\lvert\vec{c}\rvert = 25$ and $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, then find the value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$.
30.	Let θ be the angle between two unit vectors $\hat{a}$ and $\hat{b}$ such that $\sin\theta = \frac{3}{5}$. Then, $\hat{a} \cdot \hat{b}$ is equal to : (A) $\pm \frac{3}{5}$ (B) $\pm \frac{3}{4}$ (C) $\pm \frac{4}{5}$ (D) $\pm \frac{4}{3}$
31.	The vector with terminal point A (2, – 3, 5) and initial point B (3, – 4, 7) is : (A) $\hat{i} - \hat{j} + 2\hat{k}$ (B) $\hat{i} + \hat{j} + 2\hat{k}$ (C) $-\hat{i} - \hat{j} - 2\hat{k}$ (D) $-\hat{i} + \hat{j} - 2\hat{k}$
32.	If $\vec{a}$ and $\vec{b}$ are two non-zero vectors such that $(\vec{a} + \vec{b}) \perp \vec{a}$ and $(2\vec{a} + \vec{b}) \perp \vec{b}$, then prove that $\lvert\vec{b}\rvert = \sqrt{2}\,\lvert\vec{a}\rvert$.
33.	In the given figure, ABCD is a parallelogram. If $\overrightarrow{AB} = 2\hat{i} - 4\hat{j} + 5\hat{k}$ and $\overrightarrow{DB} = 3\hat{i} - 6\hat{j} + 2\hat{k}$, then find $\overrightarrow{AD}$ and hence find the area of parallelogram ABCD. 
34.	For any two vectors $\vec{a}$ and $\vec{b}$, which of the following statements is always true ? (A) $\vec{a} \cdot \vec{b} \geq \lvert\vec{a}\rvert\,\lvert\vec{b}\rvert$ (B) $\vec{a} \cdot \vec{b} = \lvert\vec{a}\rvert\,\lvert\vec{b}\rvert$ (C) $\vec{a} \cdot \vec{b} \leq \lvert\vec{a}\rvert\,\lvert\vec{b}\rvert$ (D) $\vec{a} \cdot \vec{b} < \lvert\vec{a}\rvert\,\lvert\vec{b}\rvert$

35.	The unit vector perpendicular to both vectors $\hat{i} + \hat{k}$ and $\hat{i} - \hat{k}$ is : (A) $2\hat{j}$ (B) $\hat{j}$ (C) $\frac{\hat{i} - \hat{k}}{\sqrt{2}}$ (D) $\frac{\hat{i} + \hat{k}}{\sqrt{2}}$
36.	*Assertion (A) :* For two non-zero vectors $\vec{a}$ and $\vec{b}$, $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$. *Reason (R) :* For two non-zero vectors $\vec{a}$ and $\vec{b}$, $\vec{a} \times \vec{b} = \vec{b} \times \vec{a}$.
37.	If vectors $\vec{a}$, $\vec{b}$ and $2\vec{a} + 3\vec{b}$ are unit vectors, then find the angle between $\vec{a}$ and $\vec{b}$.
38.	If $\|\vec{a}\| = 2$ and $-3 \leq k \leq 2$, then $\|k\vec{a}\| \in$: (A) $[-6, 4]$ (B) $[0, 4]$ (C) $[4, 6]$ (D) $[0, 6]$
39.	If $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + \hat{j} - \hat{k}$, then $\vec{a}$ and $\vec{b}$ are : (A) collinear vectors which are not parallel (B) parallel vectors (C) perpendicular vectors (D) unit vectors
40.	*Assertion (A) :* For any non-zero unit vector $\vec{a}$, $\vec{a} \cdot (-\vec{a}) = (-\vec{a}) \cdot \vec{a} = -1$. *Reason (R) :* Angle between $\vec{a}$ and $(-\vec{a})$ is $\frac{\pi}{2}$.
41.	$\vec{a}$, $\vec{b}$ and $\vec{c}$ are three mutually perpendicular unit vectors. If θ is the angle between $\vec{a}$ and $(2\vec{a} + 3\vec{b} + 6\vec{c})$, find the value of cos θ.
42.	Find the position vector of point C which divides the line segment joining points A and B having position vectors $\hat{i} + 2\hat{j} - \hat{k}$ and $-\hat{i} + \hat{j} + \hat{k}$ respectively in the ratio 4 : 1 externally. Further, find $\|\overrightarrow{AB}\| : \|\overrightarrow{BC}\|$.

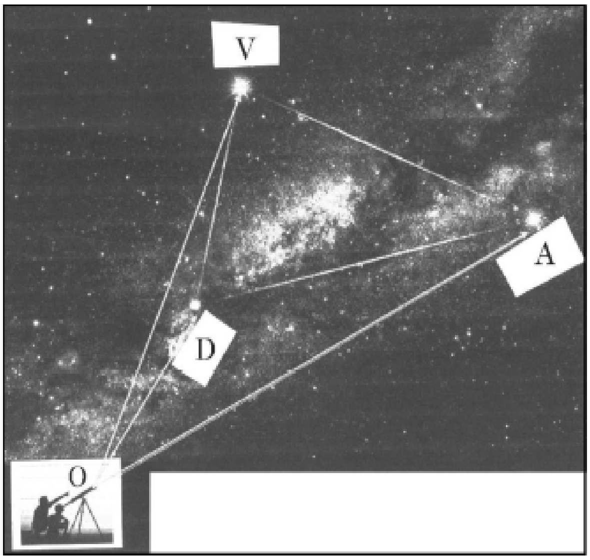
43.	If $\vec{a}$ and $\vec{b}$ are two vectors such that $\lvert \vec{a} \rvert = 1$, $\lvert \vec{b} \rvert = 2$ and $\vec{a} \cdot \vec{b} = \sqrt{3}$, then the angle between $2\vec{a}$ and $-\vec{b}$ is : (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{5\pi}{6}$ (D) $\frac{11\pi}{6}$
44.	The vectors $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - 3\hat{j} - 5\hat{k}$ and $\vec{c} = -3\hat{i} + 4\hat{j} + 4\hat{k}$ represents the sides of (A) an equilateral triangle (B) an obtuse-angled triangle (C) an isosceles triangle (D) a right-angled triangle
45.	Let $\vec{a}$ be any vector such that $\lvert \vec{a} \rvert = a$. The value of $\lvert \vec{a} \times \hat{i} \rvert^2 + \lvert \vec{a} \times \hat{j} \rvert^2 + \lvert \vec{a} \times \hat{k} \rvert^2$ is : (A) a^2 (B) $2a^2$ (C) $3a^2$ (D) 0
46.	The position vectors of points P and Q are $\vec{p}$ and $\vec{q}$ respectively. The point R divides line segment PQ in the ratio 3 : 1 and S is the mid-point of line segment PR. The position vector of S is : (A) $\frac{\vec{p} + 3\vec{q}}{4}$ (B) $\frac{\vec{p} + 3\vec{q}}{8}$ (C) $\frac{5\vec{p} + 3\vec{q}}{4}$ (D) $\frac{5\vec{p} + 3\vec{q}}{8}$
47.	**Assertion (A)** : The vectors $\vec{a} = 6\hat{i} + 2\hat{j} - 8\hat{k}$ $\vec{b} = 10\hat{i} - 2\hat{j} - 6\hat{k}$ $\vec{c} = 4\hat{i} - 4\hat{j} + 2\hat{k}$ represent the sides of a right angled triangle. **Reason (R)** : Three non-zero vectors of which none of two are collinear forms a triangle if their resultant is zero vector or sum of any two vectors is equal to the third.

48.	Find a vector of magnitude 4 units perpendicular to each of the vectors $2\hat{i} - \hat{j} + \hat{k}$ and $\hat{i} + \hat{j} - \hat{k}$ and hence verify your answer.
49.	If vector $\vec{a} = 3\hat{i} + 2\hat{j} - \hat{k}$ and vector $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, then which of the following is correct ? (A) $\vec{a} \parallel \vec{b}$ (B) $\vec{a} \perp \vec{b}$ (C) $\lvert \vec{b} \rvert > \lvert \vec{a} \rvert$ (D) $\lvert \vec{a} \rvert = \lvert \vec{b} \rvert$
50.	If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, $\lvert \vec{a} \rvert = \sqrt{37}$, $\lvert \vec{b} \rvert = 3$ and $\lvert \vec{c} \rvert = 4$, then angle between $\vec{b}$ and $\vec{c}$ is (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$
51.	The diagonals of a parallelogram are given by $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} - \hat{k}$. Find the area of the parallelogram.
52.	Two friends while flying kites from different locations, find the strings of their kites crossing each other. The strings can be represented by vectors $\vec{a} = 3\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = 2\hat{i} - 2\hat{j} + 4\hat{k}$. Determine the angle formed between the kite strings. Assume there is no slack in the strings.
53.	Find a vector of magnitude 21 units in the direction opposite to that of $\overrightarrow{AB}$ where A and B are the points A(2, 1, 3) and B(8, –1, 0) respectively.
54.	During a cricket match, the position of the bowler, the wicket keeper and the leg slip fielder are in a line given by $\vec{B} = 2\hat{i} + 8\hat{j}$, $\vec{W} = 6\hat{i} + 12\hat{j}$ and $\vec{F} = 12\hat{i} + 18\hat{j}$ respectively. Calculate the ratio in which the wicketkeeper divides the line segment joining the bowler and the leg slip fielder.

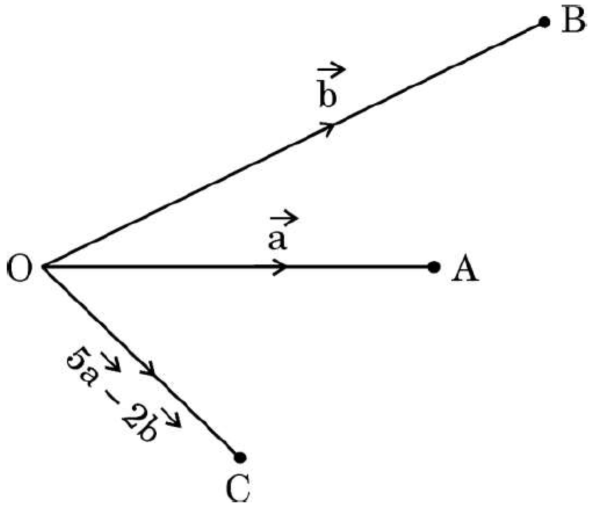
55.	The values of x for which the angle between the vectors $\vec{a} = 2x^2\hat{i} + 4x\hat{j} + \hat{k}$ and $\vec{b} = 7\hat{i} - 2\hat{j} + x\hat{k}$ is obtuse, is : (A) 0 or $\frac{1}{2}$ (B) $x > \frac{1}{2}$ (C) $\left(0, \frac{1}{2}\right)$ (D) $\left[0, \frac{1}{2}\right]$
56.	If a line makes angles of $\frac{3\pi}{4}, \frac{\pi}{3}$ and θ with the positive directions of x, y and z-axis respectively, then θ is (A) $\frac{-\pi}{3}$ only (B) $\frac{\pi}{3}$ only (C) $\frac{\pi}{6}$ (D) $\pm\frac{\pi}{3}$
57.	If $\overrightarrow{PQ} \times \overrightarrow{PR} = 4\hat{i} + 8\hat{j} - 8\hat{k}$, then the area (ΔPQR) is (A) 2 sq units (B) 4 sq units (C) 6 sq units (D) 12 sq units
58.	Let $\vec{p} = 2\hat{i} - 3\hat{j} - \hat{k}$, $\vec{q} = -3\hat{i} + 4\hat{j} + \hat{k}$ and $\vec{r} = \hat{i} + \hat{j} + 2\hat{k}$. Express $\vec{r}$ in the form of $\vec{r} = \lambda\vec{p} + \mu\vec{q}$ and hence find the values of λ and μ.
59.	A vector $\vec{a}$ makes equal angles with all the three axes. If the magnitude of the vector is $5\sqrt{3}$ units, then find $\vec{a}$.
60.	If $\vec{\alpha}$ and $\vec{\beta}$ are position vectors of two points P and Q respectively, then find the position vector of a point R in QP produced such that $QR = \frac{3}{2}QP$.
61.	If the sides AB and AC of Δ ABC are represented by vectors $\hat{j} + \hat{k}$ and $3\hat{i} - \hat{j} + 4\hat{k}$ respectively, then the length of the median through A on BC is : (A) $2\sqrt{2}$ units (B) $\sqrt{18}$ units (C) $\frac{\sqrt{34}}{2}$ units (D) $\frac{\sqrt{48}}{2}$ units

62.	Show that the area of a parallelogram whose diagonals are represented by $\vec{a}$ and $\vec{b}$ is given by $\frac{1}{2}\lvert\vec{a}\times\vec{b}\rvert$. Also find the area of a parallelogram whose diagonals are $2\hat{i}-\hat{j}+\hat{k}$ and $\hat{i}+3\hat{j}-\hat{k}$.
63.	Let $\vec{a}$ be a position vector whose tip is the point (2, – 3). If $\overrightarrow{AB}=\vec{a}$, where coordinates of A are (– 4, 5), then the coordinates of B are : (A) (– 2, – 2) (B) (2, – 2) (C) (– 2, 2) (D) (2, 2)
64.	The respective values of $\lvert\vec{a}\rvert$ and $\lvert\vec{b}\rvert$, if given $(\vec{a}-\vec{b})\cdot(\vec{a}+\vec{b})=512$ and $\lvert\vec{a}\rvert=3\lvert\vec{b}\rvert$, are : (A) 48 and 16 (B) 3 and 1 (C) 24 and 8 (D) 6 and 2
65.	Find a vector of magnitude 5 which is perpendicular to both the vectors $3\hat{i}-2\hat{j}+\hat{k}$ and $4\hat{i}+3\hat{j}-2\hat{k}$.
66.	Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three vectors such that $\vec{a}\cdot\vec{b}=\vec{a}\cdot\vec{c}$ and $\vec{a}\times\vec{b}=\vec{a}\times\vec{c}$, $\vec{a}\neq 0$. Show that $\vec{b}=\vec{c}$.
67.	A man needs to hang two lanterns on a straight wire whose end points have coordinates A (4, 1, – 2) and B (6, 2, – 3). Find the coordinates of the points where he hangs the lanterns such that these points trisect the wire AB.
68.	If $\vec{a}+\vec{b}+\vec{c}=\vec{0}$ such that $\lvert\vec{a}\rvert=3$, $\lvert\vec{b}\rvert=5$, $\lvert\vec{c}\rvert=7$, then find the angle between $\vec{a}$ and $\vec{b}$.
69.	If $\vec{a}$ and $\vec{b}$ are unit vectors inclined with each other at an angle θ, then prove that $\frac{1}{2}\lvert\vec{a}-\vec{b}\rvert=\sin\frac{\theta}{2}$.

70.	Let $\lvert \vec{a} \rvert = 5$ and $-2 \le \lambda \le 1$. Then, the range of $\lvert \lambda \vec{a} \rvert$ is : (A) [5, 10] (B) [– 2, 5] (C) [– 2, 1] (D) [–10, 5]
71.	*Assertion (A) :* If $\lvert \vec{a} \times \vec{b} \rvert^2 + \lvert \vec{a} \cdot \vec{b} \rvert^2 = 256$ and $\lvert \vec{b} \rvert = 8$, then $\lvert \vec{a} \rvert = 2$. *Reason (R) :* $\sin^2\theta + \cos^2\theta = 1$ and $\lvert \vec{a} \times \vec{b} \rvert = \lvert \vec{a} \rvert \lvert \vec{b} \rvert \sin\theta$ and $\vec{a} \cdot \vec{b} = \lvert \vec{a} \rvert \lvert \vec{b} \rvert \cos\theta$.
72.	If $\vec{a}$ and $\vec{b}$ are position vectors of point A and point B respectively, find the position vector of point C on BA produced such that BC = 3BA.
73.	Vector $\vec{r}$ is inclined at equal angles to the three axes x, y and z. If magnitude of $\vec{r}$ is $5\sqrt{3}$ units, then find $\vec{r}$.
74.	A student tries to tie ropes, parallel to each other from one end of the wall to the other. If one rope is along the vector $3\hat{i} + 15\hat{j} + 6\hat{k}$ and the other is along the vector $2\hat{i} + 10\hat{j} + \lambda\hat{k}$, then the value of λ is : (A) 6 (B) 1 (C) $\frac{1}{4}$ (D) 4
75.	If $\lvert \vec{a} + \vec{b} \rvert = \lvert \vec{a} - \vec{b} \rvert$ for any two vectors, then vectors $\vec{a}$ and $\vec{b}$ are : (A) orthogonal vectors (B) parallel to each other (C) unit vectors (D) collinear vectors
76.	The scalar product of the vector $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$ with a unit vector along sum of vectors $\vec{b} = 2\hat{i} - 4\hat{j} + 5\hat{k}$ and $\vec{c} = \lambda\hat{i} - 2\hat{j} - 3\hat{k}$ is equal to 1. Find the value of λ.

	CASE STUDY
77.	An instructor at the astronomical centre shows three among the brightest stars in a particular constellation. Assume that the telescope is located at O(0, 0, 0) and the three stars have their locations at the points D, A and V having position vectors $2\hat{i} + 3\hat{j} + 4\hat{k}$, $7\hat{i} + 5\hat{j} + 8\hat{k}$ and $-3\hat{i} + 7\hat{j} + 11\hat{k}$ respectively.  Based on the above information, answer the following questions : (i) How far is the star V from star A ? (ii) Find a unit vector in the direction of $\overrightarrow{DA}$. (iii) Find the measure of $\angle VDA$. **OR** (iii) What is the projection of vector $\overrightarrow{DV}$ on vector $\overrightarrow{DA}$?

78. Three friends A, B and C move out from the same location O at the same time in three different directions to reach their destinations. They move out on straight paths and decide that A and B after reaching their destinations will meet up with C at his predecided destination, following straight paths from A to C and B to C in such a way that $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$ and $\overrightarrow{OC} = 5\vec{a} - 2\vec{b}$ respectively.



Based upon the above information, answer the following questions :

(i) Complete the given figure to explain their entire movement plan along the respective vectors. *1*

(ii) Find vectors $\overrightarrow{AC}$ and $\overrightarrow{BC}$. *1*

(iii) (a) If $\vec{a} \cdot \vec{b} = 1$, distance of O to A is 1 km and that from O to B is 2 km, then find the angle between $\overrightarrow{OA}$ and $\overrightarrow{OB}$. Also, find $|\vec{a} \times \vec{b}|$. *2*

OR

(iii) (b) If $\vec{a} = 2\hat{i} - \hat{j} + 4\hat{k}$ and $\vec{b} = \hat{j} - \hat{k}$, then find a unit vector perpendicular to $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$. *2*